Double Trouble: Bilingual Pretraining Leaves Language-Conditioned Effects in Shared-Language Representations

Anjishnu Mukherjee Ziwei Zhu Antonios Anastasopoulos
George Mason University
{amukher6,zzhu20,antonis}@gmu.edu

Abstract

A concept can carry different associations across languages, while modern language models learn English alongside many other languages during pretraining. Yet comparisons among existing models cannot easily isolate how any one language changes the way these models represent English concepts because their training corpora, compute, architectures, and random seeds all differ. We study this question through a controlled experiment with 40 matched 310M-parameter decoder-only models that share an architecture, tokenizer, training recipe, and English data source. Each bilingual condition adds one of eight languages, while four experimental comparisons separately account for English exposure, total training, and English-document overlap. We align each model pair using 3,000 common English words, then measure where 1,000 held-out English concepts fall along 50 fixed semantic contrasts, such as red versus white. Across 32 experimental comparisons, English concept positions differ more between bilingual and English-only conditions than between English-only runs with different random seeds. These differences are larger in contextual states than in token embeddings and peak in middle layers. The language learned alongside English can therefore change how a model represents English concepts even when its English input representations are explicitly aligned[1]

1 Introduction

Language models can answer in English while drawing on representations learned from every language in their pretraining mixture. Those languages may change how English concepts are associated with one another. Consider the English concept *wedding dress*. A model trained on English and another language may place it differently along a red–white contrast than a model trained only on English, even after the two models' English token embeddings have been aligned (Figure 1). This example motivates our central question. Does learning another language alongside English change how a model organizes English concepts?

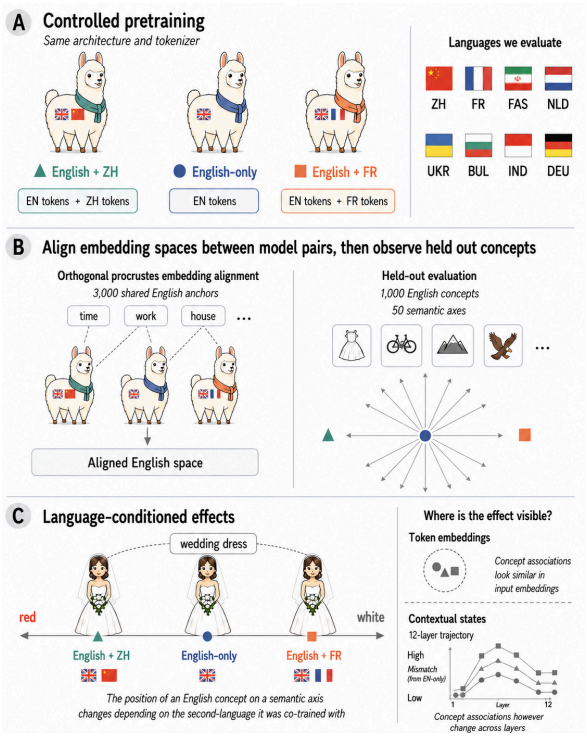


Figure 1: **Aligning English inputs can hide differences in how models represent English concepts.** A bilingual model may place *wedding dress* differently along the red–white contrast than an English-only model. We test 1,000 concepts across 50 contrasts and eight additional training languages.

Multilingual pretraining allows one model to transfer across many languages (Pires et al., 2019; Conneau et al., 2020). Shared parameters, however, do not make its representations fully language-neutral. Internal states retain information about language identity and structure (Libovický et al., 2020; Chi et al., 2020). These studies show that lan-

[1]Code available here: https://github.com/iamshnoo/bli.

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guages remain distinguishable inside multilingual models. They do not tell us whether learning one language changes relations among concepts in another. If it does, aligned English token embeddings may conceal differences in the representations built from them.

Comparisons among existing models cannot isolate this effect. Their corpora, compute budgets, tokenizers, architectures, and random seeds differ, and each difference can alter the learned representations. Linear alignment can remove arbitrary coordinate differences between independently trained embedding spaces (Mikolov et al., 2013; Smith et al., 2017; Lample et al., 2018), but a close fit on the words used to estimate a map does not establish that unseen concepts occupy the same positions. Independent training runs also differ by chance. A convincing comparison must therefore isolate language exposure, reserve new concepts for evaluation, and judge any remaining difference against same-language seed variation.

We construct this comparison with 40 matched 310M-parameter decoder-only models. The bilingual models learn English and one of eight additional languages (ZH FR FAS NLD UKR BUL IND DEU). Every run uses the same architecture, tokenizer, training recipe, and data source. We first compare models that receive the same amount of English, then models that receive the same total number of training steps. We repeat both comparisons using either the same or non-overlapping English documents. English-only runs with different random seeds measure ordinary training variation.

For each model pair, we fit an alignment using 3,000 common English words and then test 1,000 different English concepts along 50 fixed semantic contrasts. We repeat the comparison on token embeddings and on the contextual states produced at every transformer layer, fitting a new alignment at each layer. The evaluation words never influence the alignment. This design tests whether agreement on common English words extends to new English concepts rather than assuming that it does.

Across all 32 experimental comparisons, the contextual positions of held-out English concepts differ more between bilingual and English-only models than between English-only runs with different random seeds. These differences are larger than those in token embeddings and, on average, largest in middle layers. Figure 2 shows why direction also matters. The same English concept can move toward opposite ends of a semantic contrast depending on the language learned alongside English. The central result is therefore not merely a gap between embeddings and contextual states. In our controlled models, the additional training language changes where English concepts fall relative to the same semantic contrasts, even after the representations have been explicitly aligned.

The paper makes three contributions.

- **We isolate the effect of learning an additional language.** We train 40 models in which each bilingual model learns English and one of eight other languages. Every model uses the same architecture, tokenizer, data source, and training recipe. The comparisons either match English exposure or total training steps and use shared or non-overlapping English documents. English-only runs with different random seeds establish how much independently trained models normally differ.
- **We evaluate held-out English concepts after alignment.** We align each model pair using 3,000 common English words, then measure where 1,000 different English concepts fall along 50 fixed semantic contrasts, such as red versus white. We perform this comparison on token embeddings and on contextual states at every transformer layer.
- **We find that the additional language changes how English concepts are related.** Across 32 experimental comparisons, bilingual models place held-out English concepts differently from matched English-only models by more than English-only runs differ from one another. These differences are larger in contextual states than in token embeddings and largest in middle layers on average. Different training languages can also move the same concept toward opposite ends of a contrast.

2 Related Work

Our study connects work on aligning independently trained representations, distinguishing shared from language-specific information, and measuring associations among concepts. Together, these methods let us ask whether an additional training language changes how a model represents English after alignment.

Comparing independently trained representations. Linear mappings between monolingual embedding spaces are widely used for bilingual lexicon induction, translation, and nearest-neighbor



Figure 2: **The same English concept can move toward either end of a semantic contrast depending on the second training language.** Each horizontal line names one contrast, and the colored points show all eight bilingual conditions. The plotted value is the bilingual projection minus the aligned English-only projection. Points to the left move toward the first endpoint, and points to the right move toward the second. These twelve examples show the directional reversals. The aggregate results use all 1,000 concepts and 50 contrasts.

retrieval (Schönemann, 1966; Mikolov et al., 2013; Smith et al., 2017; Lample et al., 2018; Artetxe et al., 2018). Their accuracy depends on the languages, domains, and embedding methods being compared (Søgaard et al., 2018; Vulić et al., 2020; Patra et al., 2019). Related methods support zero-shot translation (Mullov et al., 2024) and multilingual in-context learning (Li et al., 2024). We use alignment to remove arbitrary rotations between two independently trained English representation spaces. The fitted transformation places the spaces in comparable coordinates, but the 1,000 concepts used for evaluation are not among the 3,000 words used to estimate it. We also fit a separate transformation at every transformer layer. Agreement on the alignment words therefore does not determine the result on the held-out concepts.

Multilingual models. Pretraining enables cross-lingual transfer (Pires et al., 2019; Conneau et al., 2020). The resulting representations still contain language-specific information (Libovický et al., 2020; Chi et al., 2020). Prior work has compared multilingual and monolingual representations (Zhao et al., 2023), separated language-sensitive components from shared components (Chang et al., 2022), recovered typological information (Choenni and Shutova, 2022), and traced shared features as they become language-specific outputs (Harrasse et al., 2025). These studies show that language identity remains visible inside multilingual models. They cannot isolate the effect of adding one language because the models being compared also differ in data, training, or architecture. Our matched models isolate that question. We also vary English-document overlap because repeated documents can change memorization and evaluation results (Dodge et al., 2021; Lee et al., 2022). Shared parameters and related writing systems may strengthen cross-lingual transfer (Dufter and Schütze, 2020; Muller et al., 2021), but our question is whether the added language also changes relations among English concepts.

Interpreting concept associations. Directions between contrasting words have been used to study bias and historical change in word embeddings (Bolukbasi et al., 2016; Caliskan et al., 2017; Garg et al., 2018). Related cross-cultural work examines value associations in pretrained models (Arora et al., 2023). We use each contrast, such as *individual* versus *collective*, as a fixed reference for comparing the same English concept across training conditions. These contrasts do not assign a score to a language, culture, or speaker. Research on cross-cultural values motivated the nine broad themes (Hofstede, 2001; Schwartz, 2006; Inglehart and Welzel, 2005; House et al., 2004), but we wrote the endpoint pairs and fixed them before analysis. We retain the direction of each change because an average absolute difference would hide cases in which different training languages move the same concept toward opposite endpoints.

3 Experimental Design

Each experiment compares an English-only model with a bilingual model after aligning their English representations. The models share an architecture, tokenizer, training recipe, and data source. Separate comparisons account for English exposure, total training, English-document overlap, and random initialization. Held-out concepts test whether the alignment extends beyond the words used to estimate it.

3.1 What We Measure

For each English-only model M_{EN} and matched bilingual model $M_{\text{EN+L2}}$, we measure disagreement on held-out English concepts after alignment. Pairs of English-only models trained with different random seeds provide the reference. For any disagreement measure m, we report

$$\Delta m \;=\; m(M_{\text{EN}}, M_{\text{EN+L2}}) \;-\; \bar{m}_{\text{seed-var}},$$

where $\bar{m}_{\text{seed-var}}$ is the average disagreement between English-only runs that differ only in random seed. A value near zero means that the bilingual and English-only models differ by about as much as two English-only runs. Positive values indicate disagreement beyond this same-language reference.

3.2 Training and Controls

Data and languages. BabyBabelLM provides comparable 100-million-token collections for eight non-English languages (Jumelet et al., 2026). We use Chinese, French, Persian, Dutch, Ukrainian, Bulgarian, Indonesian, and German (ZH FR FAS NLD UKR BUL IND DEU). Every run uses the same 310M-parameter decoder-only architecture and training recipe. Under this recipe, 1,500 steps process about 50 million tokens and 3,000 steps process about 100 million tokens. The appendix gives the complete architecture, batching, and tokenizer details.

What we hold fixed. Three training differences could otherwise explain the result. A bilingual model may see less English, receive more training, or share documents with its English-only counterpart. We test each explanation directly.

- ***same English exposure.*** The English-only model trains for 1,500 steps. The bilingual model trains for 3,000 steps, alternating equal blocks of English and the additional language. Both models therefore receive exactly 1,500 English steps.
- ***same total training steps.*** Both models train for 3,000 steps. The English-only model receives only English, while the bilingual model receives 1,500 English steps and 1,500 steps in the additional language.

We repeat both training comparisons under two document conditions.

- **Shared English documents.** Both models train on the same English documents.
- **Separate English documents.** The models train on non-overlapping English documents.

Table 1 shows the resulting four comparisons and the explanation tested by each one.

The same-English-exposure comparison with separate documents is the strictest control for English data. Both models receive 1,500 English steps drawn from non-overlapping documents. Only the bilingual model receives another 1,500 steps in the additional language. The matched-total-training comparisons address a different explanation by giving both models the same total number of steps.

3.3 Evaluation Concepts and Semantic Contrasts

We estimate each alignment using 3,000 common English words selected by frequency and filtered to remove stop words, non-ASCII text, and overlap with the evaluation sets. A separate set of 1,000 English concepts covers values and norms, family and kinship, religion and ritual, food and cuisine, festivals and holidays, clothing and appearance,

Comparison	Question answered	English steps	Total training steps	English documents	Median contextual ΔD_{Axis}
Same English exposure	Does a difference remain when both models see the same amount of English?	$(1500, 1500)$	$(1500, 3000)$	shared	**0.16**
Same English exposure, separate documents	Does the difference remain with no English documents in common?	$(1500, 1500)$	$(1500, 3000)$	non-overlapping	**0.18**
Same total training steps	Does a difference remain after matching the number of training steps?	$(3000, 1500)$	$(3000, 3000)$	shared	**0.53**
Same total training steps, separate documents	Does the difference remain without shared English documents?	$(3000, 1500)$	$(3000, 3000)$	non-overlapping	**0.32**

Table 1: **The four comparisons test English exposure, total training, and English-document overlap.** Each numerical pair lists the English-only model first and the bilingual model second. Every bilingual model receives 1,500 English steps and 1,500 steps in the additional language. The English-only model receives either 1,500 steps to match English exposure or 3,000 steps to match total training. The median contextual difference exceeds English-only seed variation in all four comparisons.

symbols and colors, governance and law, social identity, and daily customs.

Each of the 50 semantic contrasts pairs two endpoints, such as *individual* and *collective*. The contrasts span nine themes motivated by research on cross-cultural values (Hofstede, 2001; Schwartz, 2006; Inglehart and Welzel, 2005; House et al., 2004) and prior cross-cultural NLP analysis (Arora et al., 2023). We wrote the endpoint pairs ourselves and fixed them before examining the results. They provide common reference directions for comparing concepts, not scores assigned to a language, culture, or speaker. Table 3 gives one example from each theme.

We also evaluate 100 common physical terms, including body parts, adapted from basic-vocabulary traditions (Swadesh, 1955). These negative controls test whether the difference is limited to the primary concept categories. For the non-English extension, we translate the words with NLLB (NLLB Team et al., 2022), score translation quality with COMETKiwi (Rei et al., 2022), back-translate them, and manually review low-confidence cases. The primary analysis uses only the original English words. Complete word lists and translation checks appear in Appendix A.2.

3.4 Alignment and Metrics

For each model pair (A, B), token embeddings are compared only with token embeddings, and contextual states are compared only with contextual states. Let $X^a, X^b \in \mathbb{R}^{V \times d}$ denote either representation. We use the English alignment words $\mathcal{N}$ to fit an orthogonal Procrustes transformation.

$$W^\star = \arg\min_{W^\top W = I} \|X^a_{\mathcal{N}} W - X^b_{\mathcal{N}}\|_F.$$

This transformation makes the coordinate systems comparable without changing distances or angles within either space. We then measure disagreement on the held-out concepts $\mathcal{C}$. Our primary measure, **axis projection disagreement**, asks whether the same concept occupies the same relative position along a semantic contrast after alignment.

$$D_{Axis,i} = \frac{1}{|\mathcal{C}|} \sum_{w \in \mathcal{C}} \left| \langle x^a_w, \hat{u}^a_i \rangle - \left\langle x^b_w, \hat{u}^b_i \right\rangle \right|,$$

for contrast i. After applying the fitted transformation to model A, x^a_w and x^b_w are the aligned representations of concept w. The unit vectors $\hat{u}^a_i$ and $\hat{u}^b_i$ point from the first endpoint of the contrast to the second in each model. The notation $\langle \cdot, \cdot \rangle$ denotes the Euclidean inner product.

D_{Axis} is the absolute difference between the two projections. For examples such as Figure 2, we retain the sign to show which endpoint the bilingual representation moves toward relative to the English-only representation. Two complementary measures compare each concept's 25 nearest neighbors, denoted by D_{NN}, and the pairwise cosine similarities among concepts, denoted by D_{Struct}. Experiment 4 also tests a less constrained affine transformation.

Every reported Δ subtracts the English-only seed mean defined above. Positive values therefore exceed the average disagreement between English-only runs. At each checkpoint, this reference uses six ordered comparisons corresponding to three unique seed pairs.

4 Results

Unequal English exposure, longer training, or reused documents could each create an apparent language effect. Experiments 1 and 2 test these alternatives. Experiments 3 and 4 determine where the remaining difference appears across layers and checkpoints and whether it survives other alignment methods.

4.1 Experiment 1. Does the difference remain after matching English exposure or total training?

If either unequal English exposure or unequal total training causes the difference, matching the relevant quantity should reduce it to ordinary seed variation. The first comparison gives both models the same number of English steps. The second gives both models the same total number of training steps.

Figure 3 shows all eight additional languages under the four comparisons in Table 1. Every contextual value exceeds the English-only seed mean, although some language-level intervals are wide. The difference remains when the paired models receive the same amount of English and train on the same English documents. Less English and different English text are therefore not sufficient explanations.

> **Takeaway 1.** Contextual differences remain above English-only seed variation when either English exposure or total training is matched.

4.2 Experiment 2. Is the difference driven by shared English documents?

Training on the same English documents could make two models appear similar for reasons unrelated to language exposure. We therefore repeat every comparison with non-overlapping English documents. Removing shared documents does not consistently increase or decrease the contextual difference across languages (Figure 4). Changes in token embeddings remain close to zero. The contextual difference still exceeds English-only seed variation for most language conditions, with language-level uncertainty reported in Appendix Figure 9.

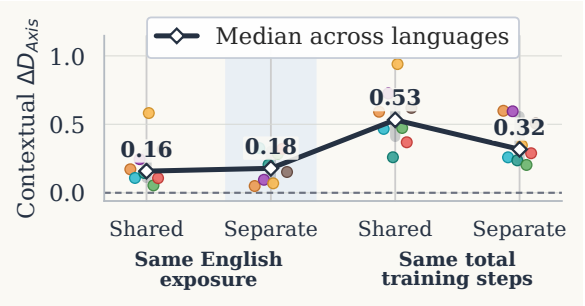


Figure 3: **The contextual difference exceeds English-only seed variation in all four comparisons.** Each point represents one additional language, and the dark line shows the median. The four positions match either English exposure or total training and use shared or separate English documents. Zero is the average difference between English-only runs.

> **Takeaway 2.** Sharing English documents does not account for the difference between bilingual and English-only models.

4.3 Experiment 3. Where in the model is the difference largest?

Aligning token embeddings constrains only the model input. Each transformer layer can still change how a concept is represented in context. We therefore fit a separate alignment to the contextual states from each of the 12 layers. We also normalize every vector to unit length and standardize projections using neutral English words. These two checks test whether vector length alone produces the layerwise pattern.

For every additional language, contextual states differ more than token embeddings when the concepts are evaluated in English (Figure 5a). The same ordering holds when concepts are evaluated in the additional language (Figure 5b). The signed examples in Figure 2 show that the aggregate difference can arise from movement in either direction along a contrast.

The average difference is small at the aligned input layer, rises through the middle transformer layers, and decreases in the final layers while remaining above its input value (Figure 6a). Because we fit a new alignment at every layer, this rise cannot be attributed only to using an input-layer transformation on later states. Normalizing vector lengths or standardizing projections against neu-

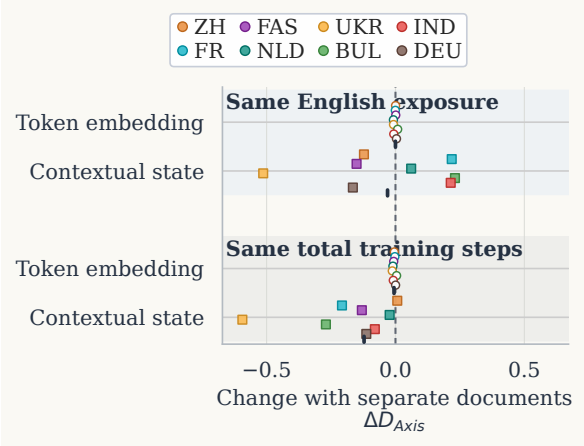


Figure 4: **Removing shared English documents does not consistently reduce the contextual difference.** Each point shows the change after replacing shared English documents with non-overlapping sets. Token-embedding changes remain near zero, while contextual changes vary in direction across languages.

tral English words preserves the middle-layer peak (Figure 6b). Vector length alone therefore does not explain the result.

> **Takeaway 3.** English concept positions differ most in middle layers on average, even after controlling for vector length.

4.4 Experiment 4. Does the difference appear early and remain under other alignments?

If the difference emerges only at the end of training, it should be absent from earlier checkpoints. We therefore evaluate every saved checkpoint at intervals of 500 steps. We also fit the orthogonal transformation directly to contextual states and test a less constrained affine transformation. If the original alignment creates the difference, these alternatives should substantially reduce it. Every transformation is evaluated on concepts that were not used for alignment (Table 2).

The contextual difference appears at the first 500-step checkpoint and remains at every saved checkpoint through 3,000 steps (Figure 7). On held-out concepts, mean ΔD_{Axis} is 0.386 whether the orthogonal transformation is fitted to token embeddings or contextual states. The affine transformation yields 0.580 (Table 2). Changing the subset of English words used for alignment also preserves the separation between token embeddings and contextual states (Appendix Table 13). The result is therefore neither confined to the final checkpoint nor removed by a more flexible alignment.

> **Takeaway 4.** The difference appears early and remains when the orthogonal transformation is fitted directly to contextual states or replaced with an affine transformation.

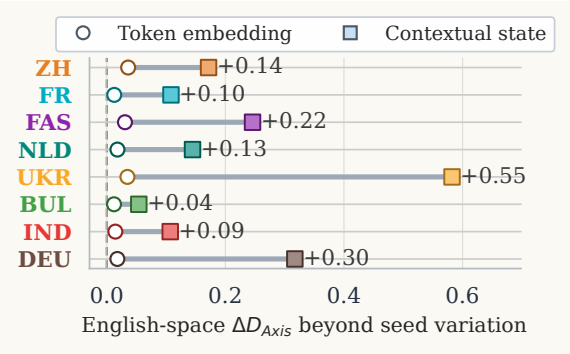
5 Discussion

What alignment shows. Aligning common English words shows that two independently trained models can express those words in comparable coordinates. It does not show that other concepts retain the same relationships. The representations of our held-out concepts remain different even when the transformation is fitted directly to contextual states. Prior work has also found that multilingual representations change across layers (Chi et al., 2020; Wendler et al., 2024). A layer-level comparison should therefore fit the alignment at that layer, evaluate different concepts, and compare the remaining disagreement with same-language training variation.

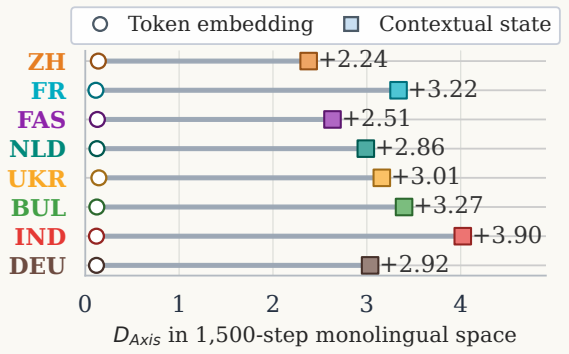
How to interpret the directional changes. Figure 2 places all eight language conditions between explicitly named endpoints. The same concept can move toward either endpoint depending on the additional training language. A single ranking would hide this result. The absolute measure summarizes how far the models disagree, while the signed examples show the direction of individual changes. The contrasts listed in Table 3 are fixed references for concepts, not scales for ranking languages or cultures.

What the training controls rule out. The four comparisons address different explanations. Matching English exposure shows that the result is not explained only by the bilingual model seeing less English. Matching total training shows that it is not explained only by additional training steps. Using non-overlapping English documents shows that repeated English text is not required. These controls do not identify the mechanism. Cross-language differences in word meaning, co-occurrence, or the way related meanings compete for the same model parameters may all contribute. The middle-layer result is consistent with contextual processing strengthening the difference, but it does not distinguish among these explanations.

Where translation quality matters. The main analysis uses the original English concepts and does not depend on translation. Translation is required only when we repeat the evaluation in the ad-

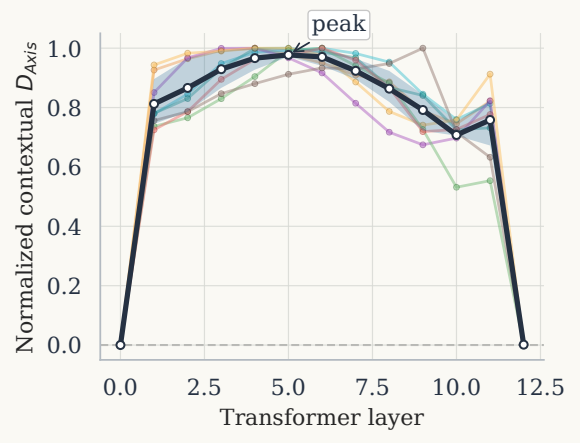


(a) In English, contextual states differ more than token embeddings for every additional training language.

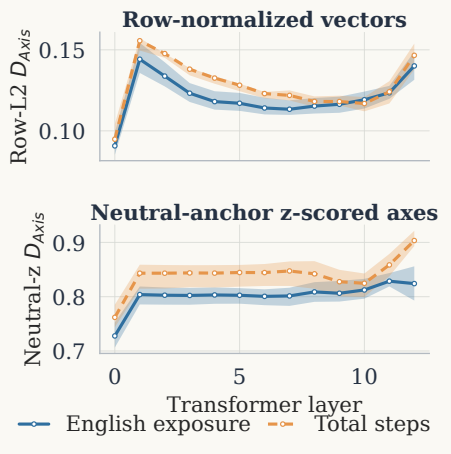


(b) The same ordering holds when evaluation is performed in each additional language.

Figure 5: **Contextual states differ more than token embeddings in both English and the additional language. (a)** In English, squares show contextual states and circles show token embeddings for each additional training language. **(b)** Evaluating concepts in each additional language yields the same ordering.



(a) Difference across transformer layers.



(b) Differences across layers after controlling for vector length.

Figure 6: **Differences in English concept placement are largest in middle layers on average. (a)** Thin lines show the eight additional languages, the dark line shows their mean, and the band spans the 20th to 80th percentiles. **(b)** Normalizing vector lengths and standardizing against neutral English words preserve the middle-layer peak.

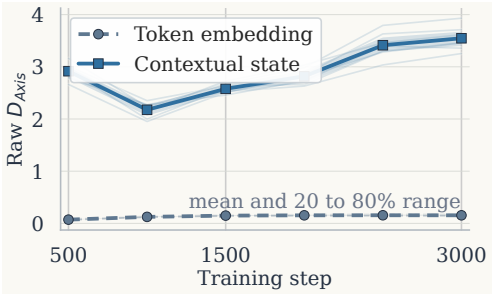


Figure 7: **Differences in English concept placement appear at the first saved checkpoint and persist through training.** Lines average the eight additional languages under matched English exposure with shared English documents. Checkpoints are 500 steps apart. Bands span the 20th to 80th percentiles across languages.

ditional language (Figure 5b). Mean COMETKiwi scores range from 0.745 to 0.797, and the proportion scoring at least 0.80 ranges from 52.9% to 71.7% (Table 4). Across these quality groups, replacing shared English documents with separate sets changes median contextual neighbor overlap by at most 0.028. Translation quality is therefore unlikely to explain the document-overlap result, although individual translations still warrant further review.

Connection to prior work. Wendler et al. (2024) found that Llama-2's intermediate states become more similar to English before later states move toward the language of the input. That work explains how several languages coexist within one multilingual model. We ask a complementary ques-

Representation	Alignment	Fitted on	Held-out difference (mean ± SD) D_{NN}	D_{Struct}	D_{Axis}	Alignment fit Error / word
Token embedding	Orthogonal	Token embedding	0.029 ± 0.019	0.014 ± 0.015	0.014 ± 0.012	0.041
	Orthogonal	Contextual state	0.029 ± 0.019	0.014 ± 0.015	0.014 ± 0.012	0.194
	Affine	Token embedding	0.097 ± 0.026	0.055 ± 0.024	0.021 ± 0.023	0.030
Contextual state	Orthogonal	Token embedding	0.015 ± 0.016	0.042 ± 0.038	0.386 ± 0.248	0.041
	Orthogonal	Contextual state	0.015 ± 0.016	0.042 ± 0.038	0.386 ± 0.248	0.194
	Affine	Token embedding	0.027 ± 0.016	0.052 ± 0.043	0.580 ± 0.260	0.030

Table 2: **The difference remains when alignment is fitted directly to contextual states.** Across sixteen shared-document comparisons, contextual ΔD_{Axis} is 0.386 under either orthogonal fit and 0.580 with an affine transformation. English-only seed variation is subtracted, and every concept is held out from alignment.

Theme	Endpoint 1	Endpoint 2	Count
Values and norms	individualism	collectivism	10
Family and kinship	mother	father	5
Religion and ritual	sacred	secular	5
Governance and law	democracy	monarchy	5
Food and cuisine	rice	bread	5
Festivals and holidays	fasting	feasting	5
Clothing and appearance	veiled	unveiled	5
Symbols and colors	white	red	5
Social identity	local	global	5

Table 3: **The 50 fixed semantic contrasts cover nine themes.** One example and the number of contrasts in each theme are shown. Appendix Tables 5 and 6 give the full list and sources.

Language	Mean / ≥ 0.80 score (%)	Review / duplicate (%)	Largest median overlap change
BUL	0.797 / 71.7	37.1 / 4.7	0.021
DEU	0.779 / 66.6	35.1 / 4.7	0.011
FAS	0.797 / 69.6	51.0 / 11.8	0.021
FR	0.777 / 65.8	28.8 / 3.2	0.020
IND	0.764 / 59.9	38.9 / 9.0	0.000
NLD	0.758 / 58.3	35.1 / 5.5	0.021
UKR	0.791 / 69.5	32.8 / 6.0	0.028
ZH	0.745 / 52.9	49.1 / 13.1	0.000

Table 4: **Translation quality varies, but the document-overlap result is similar across quality groups.** The final column gives the largest absolute median change in contextual neighbor overlap after using non-overlapping English documents.

tion across matched models. What changes within English when another language is added to training? Our results show that the added language changes measurable relations among English concepts. Multilingual training can therefore retain language-specific information while also changing representations within English.

Implications for model comparison. Two models can align closely on 3,000 common English words while disagreeing on 1,000 different concepts. An alignment score on its own is therefore insufficient evidence that the models represent English similarly. The comparison should be made at the layer being interpreted, using concepts that did not determine the alignment. Reporting both direction and magnitude distinguishes opposing changes from a common shift, and English-only runs reveal whether the disagreement exceeds ordinary training variation.

Open questions. Our 310M-parameter decoder-only models provide a controlled first measurement at one scale. The same held-out comparison should be repeated with larger architectures, mixtures containing several additional languages, and multiple seeds for every bilingual condition. More languages are also needed before testing whether script or language family predicts the direction or size of a change. Finally, output evaluations must determine when a difference in internal representations changes generated text.

6 Conclusion

Agreement on familiar English words does not imply agreement on new concepts. In our controlled comparisons, an additional training language changed where held-out English concepts fell along fixed semantic contrasts. The difference exceeded English-only seed variation, varied in direction across languages, and peaked in middle layers on average. Comparisons should therefore fit and evaluate alignments on separate concepts at the relevant layer and use same-language runs as a reference. More broadly, a pretraining mixture can shape relations within English even when English input representations appear aligned.

Limitations

We study one 310M-parameter decoder-only architecture so that training conditions can be closely matched. The evidence therefore does not establish that the same pattern holds at larger scales or in other architectures. Each bilingual condition uses one random seed. English-only runs estimate ordinary seed variation, but repeated bilingual runs are needed to measure uncertainty within each language condition. The evaluation also depends on 50 hand-authored semantic contrasts and, for the non-English extension, translated concepts. These contrasts provide fixed reference directions rather than a complete account of any language or culture. We measure internal representations, not generated behavior, so the results do not show that the observed differences change model outputs.

Ethical Considerations

This study analyzes model representations and neither deploys a system nor uses personal or user-generated data. The main risk lies in how the results are interpreted. A difference associated with one training language must not be treated as a property of that language, its cultures, or its speakers. Every language contains substantial regional, social, and historical variation, while our training data and hand-authored contrasts capture only narrow samples. The contrasts are tools for comparing models, not scales for ranking languages or people.

We report every evaluated language rather than selecting only large effects. We retain the direction of individual changes, compare them with English-only seed variation, and include common physical terms as negative controls. We also distinguish the primary English analysis from the translated extension and report translation-quality checks. These choices make the scope of the evidence clearer, but they do not make the concepts culturally comprehensive or guarantee every translation.

The results concern internal representations, not generated behavior. They should not be used to infer a speaker's identity, evaluate the quality of a language, or justify removing a language from training. Instead, they motivate direct evaluation of model outputs. Before drawing conclusions about deployment, developers should test whether differences among internal English concept representations produce corresponding differences in generated text.

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Appendix

Appendix A.1 records implementation details, Appendix A.2 gives the probe lists and translation checks, and Appendix A.3 records the data and evaluation split. Appendix A.4 organizes the additional results by the question each analysis answers.

A.1 Complete Implementation Details

All runs use one Llama-style decoder with pre-norm RMSNorm and approximately 310M parameters. It has 12 layers, hidden size 768, FFN size 3072, 12 attention heads, and untied token/output matrices. Every checkpoint uses the unchanged `meta-llama/Llama-3.2-1B` BPE tokenizer and its complete 128,256-token multilingual vocabulary; we train one shared tokenizer and retain its full vocabulary in every condition. Section A.4.4 reports the tokenizer-identity audit. Throughput is fixed at 32,768 tokens per step, using batch size 8, gradient accumulation 8, and sequence length 512. For non-English probe construction, we use NLLB for translation, COMETKiwi for quality scoring, back-translation, and manual review of low-confidence cases.

A.2 Probe Details

Semantic Axes Table 5 lists every axis, and Table 6 gives the literature that motivated each theme. The endpoint pairs are hand-authored rather than taken verbatim from the cited sources, and they were fixed before analysis.

Probe Translation and QC Summary Table 7 reports translation checks by language. The corresponding result by quality tier appears in Section A.4.

A.3 Data Statistics and Evaluation Split

Table 8 records the data, model, and evaluation split used throughout the study.

Item	Statistics / split detail
Pretraining data source	BabyBabelLM English and 8 Tier-1 non-English languages used in this study.
Target-language inventory	8 languages spanning ●ZH ●FR ●FAS ●NLD ●UKR ●BUL ●IND ●DEU.
Model configuration	Llama-style pre-norm decoder with RMSNorm, 12 layers, hidden size 768, FFN size 3072, 12 attention heads, vocabulary size 128,256, and untied token-embedding/output matrices (∼310M parameters).
Token budget checkpoints	1500 steps (≈49.15M tokens) and 3000 steps (≈98.30M tokens), at 32,768 tokens/step.
EN-centered comparisons	16 primary pairs (2 EN reference lengths × 8 EN+L2 conditions); shared/separate-document conditions give 32 model-pair observations. Token-embedding/contextual-state results are two views of these pairs; English-only seed variation uses six ordered comparisons (three unique seed pairs) per checkpoint.
General-vocabulary anchors (alignment set)	3000 English words; used only to estimate alignment maps.
Semantic probes (evaluation set)	1000 terms; held out from alignment and used only to measure model differences.
Semantic axes	50 axis pairs; fixed before training/evaluation and used only in evaluation metrics.
Negative-control set	100 common physical-vocabulary words used only as an evaluation check.
Evaluation split	This unsupervised pretraining study has no supervised train/dev/test split. General-vocabulary words are used for alignment; separate semantic probes, axes, and negative controls are used only for evaluation.

Table 8: Data statistics and evaluation split. Words used for alignment are separate from the words used for evaluation.

Corpus and language-mixing check. BabyBabelLM provides language-specific mixtures of transcribed child–caregiver and adult conversations, educational material, children's books, child-oriented Wikipedia and news, and child-suitable film/television subtitles; where needed, it pads Tier-1 budgets with filtered OpenSubtitles and, for some languages, FineWeb-C or Wikipedia. All training data come from these BabyBabelLM source pools. Category proportions vary with language-specific source availability, so cross-language magnitudes remain descriptive. Within each target-language

Axis set A			Axis set B	
#	Endpoint 1	Endpoint 2	# Endpoint 1	Endpoint 2
1	individualism	collectivism	26 rice	bread
2	hierarchy	equality	27 tea	coffee
3	duty	freedom	28 spicy	mild
4	honor	shame	29 vegetarian	meat
5	obedience	autonomy	30 fermented	fresh
6	tradition	modernity	31 fasting	feasting
7	restraint	indulgence	32 lunar	solar
8	conformity	dissent	33 mourning	celebration
9	solidarity	competition	34 ancestor	novelty
10	certainty	ambiguity	35 pilgrim	spectator
11	mother	father	36 veiled	unveiled
12	elder	youth	37 formal	casual
13	kin	stranger	38 ornate	plain
14	lineage	mobility	39 modest	revealing
15	clan	individual	40 uniform	personalized
16	sacred	secular	41 white	red
17	ritual	routine	42 black	gold
18	prayer	reason	43 dragon	eagle
19	pilgrimage	tourism	44 lotus	rose
20	taboo	permissive	45 script	image
21	democracy	monarchy	46 local	global
22	federal	centralized	47 indigenous	diaspora
23	customary	codified	48 assimilation	pluralism
24	authority	deliberation	49 majority	minority
25	patronage	meritocracy	50 homeland	migration

Table 5: Complete list of 50 semantic axes. The same fixed contrasts are reused in every language setting, and no endpoint pair is fitted to the reported results.

#	Endpoint 1	Endpoint 2	Category	Citation(s)
1	individualism	collectivism	Values and social norms	(Hofstede, 2001; Schwartz, 2006)
2	hierarchy	equality	Values and social norms	(Hofstede, 2001; Schwartz, 2006)
3	duty	freedom	Values and social norms	(Inglehart and Welzel, 2005)
4	honor	shame	Values and social norms	(House et al., 2004)
5	obedience	autonomy	Values and social norms	(Schwartz, 2006)
6	tradition	modernity	Values and social norms	(Inglehart and Welzel, 2005)
7	restraint	indulgence	Values and social norms	(Hofstede, 2001)
8	conformity	dissent	Values and social norms	(Schwartz, 2006)
9	solidarity	competition	Values and social norms	(House et al., 2004)
10	certainty	ambiguity	Values and social norms	(Hofstede, 2001)
11	mother	father	Family and kinship	(House et al., 2004)
12	elder	youth	Family and kinship	(Hofstede, 2001)
13	kin	stranger	Family and kinship	(Schwartz, 2006)
14	lineage	mobility	Family and kinship	(Inglehart and Welzel, 2005)
15	clan	individual	Family and kinship	(Schwartz, 2006)
16	sacred	secular	Religion and ritual life	(Inglehart and Welzel, 2005)
17	ritual	routine	Religion and ritual life	(House et al., 2004)
18	prayer	reason	Religion and ritual life	(Inglehart and Welzel, 2005)
19	pilgrimage	tourism	Religion and ritual life	(Hershcovich et al., 2022)
20	taboo	permissive	Religion and ritual life	(Hofstede, 2001)
21	democracy	monarchy	Governance, institutions, and law	(Hofstede, 2001)
22	federal	centralized	Governance, institutions, and law	(House et al., 2004)
23	customary	codified	Governance, institutions, and law	(Hofstede, 2001)
24	authority	deliberation	Governance, institutions, and law	(House et al., 2004)
25	patronage	meritocracy	Governance, institutions, and law	(Hofstede, 2001)

Table 6: Theme-level sources for all 50 semantic axes. The cited works motivate broad domains, not the exact hand-authored endpoint pairs.

Lang	n	Mean QE	Median QE	High	Medium	Low	Manual-review	Duplicate-target
BUL	1000	0.796	0.847	717	192	91	371	47
DEU	1000	0.779	0.824	666	242	92	351	47
FAS	1000	0.797	0.833	696	249	55	510	118
FR	1000	0.777	0.832	658	227	115	288	32
IND	1000	0.763	0.820	599	262	139	389	90
NLD	1000	0.758	0.820	583	270	147	351	55
UKR	1000	0.791	0.841	695	210	95	328	60
ZH	1000	0.745	0.806	529	312	159	491	131

Table 7: Translation checks for probes in all non-English languages. QE uses COMETKiwi when available (high $\geq$ 0.80, medium [0.60, 0.80), low $<$ 0.60); the table also reports manual-review flags and duplicate translations.

experiment, all conditions use the same data preparation procedure, and shared/separate conditions partition the same English pool.

Following Shao et al. (2026), we also ran a heuristic chunk-level fastText audit over all 3,213,686 documents used here. We flag a mixed-language candidate when the target and another language each cover at least 20% of high-confidence sampled segments, with at least three such segments. This identifies 12,382 candidates (0.385%). Among the 3,075,976 non-English documents, 2,721 (0.088%) are target-plus-English candidates; per-language rates range from 0.001% to 1.217%. Shao et al. (2026) found that bilingual documents comprised about 2% of the 240B-token FineWeb mixtures they studied, using a different corpus and filter. We treat sparse mixed-language exposure as one candidate mechanism for future intervention studies.

Fixed training recipe and reported values. One pre-specified training recipe is reused for every run, keeping tuning choices fixed across conditions. All runs use the same 310M-parameter model configuration reported in Section 3.2, batch size 8, gradient accumulation 8, sequence length 512, and throughput 32,768 tokens/step, with checkpoints at 1500 and 3000 steps.

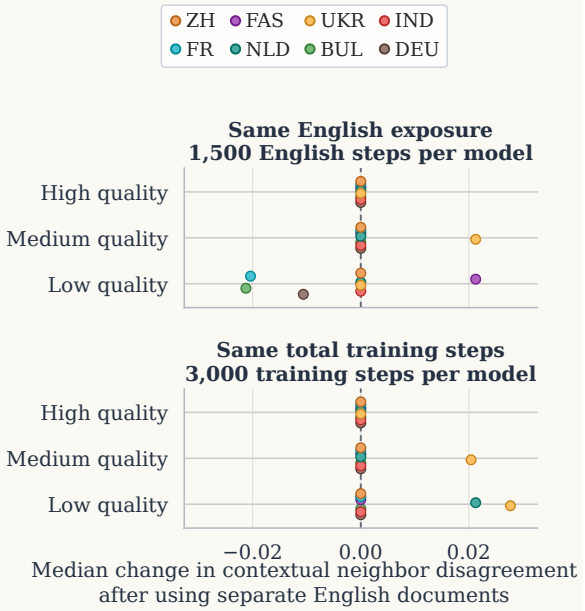


Figure 8: **The document-overlap pattern is stable across translation-quality tiers.** Each point is one additional language. The upper panel holds English exposure fixed at 1,500 English training steps per model; the lower panel holds total training fixed at 3,000 steps per model. Values near zero indicate similar contextual neighbor disagreement after replacing shared English documents with separate sets.

A.4 Additional Results

A.4.1 Translation-Quality Pattern

Replacing shared English documents with separate sets changes contextual neighbor disagreement by similar amounts across translation-quality tiers. Figure 8 shows the pattern for all eight languages and both training controls.

A.4.2 How to Read the Robustness Evidence

The following results are organized by the question a reader may want to check. Table 9 points to

the most informative display for each claim. Figure 9 then shows all language-level intervals for the primary metric and all four training controls.

A.4.3 Matched Total Training and Same-Language Controls

The main figures emphasize the setting in which the English-only and bilingual models receive the same amount of English. The following displays isolate the complementary setting in which both models instead receive 3,000 total training steps. Here the English-only model receives all steps in English, while the bilingual model receives 1,500 English steps and 1,500 steps in the additional language.

Same-language controls. Six ordered English-only comparisons, corresponding to three unique seed pairs, define the seed-variation mean used in the English-centered analyses.

Axis-group holdout. This analysis tests whether one axis group drives the contextual difference. Both matched and held-out groups retain positive seed-centered values.

A.4.4 Robustness Beyond the Main Controls

Training-progress sensitivity. These results ask whether the difference between contextual states and token embeddings appears only late in training. Across all eight language conditions, it is present at the earliest saved checkpoint and persists throughout training.

Alternative alignment maps. Switching to an affine map changes the values but does not bring contextual D_{Axis} close to English-only seed variation.

Anchor words. The contextual-state difference is stable when the alignment map is fitted using random subsets of 500 to 3,000 English anchor words.

Nearest-neighbor parameter sensitivity. Table 14 varies the neighborhood size used by the complementary D_{NN} metric. Mean centered disagreement remains positive across all tested values of k, although the ordering of embeddings and contextual states varies with k. Our main comparison therefore rests on the primary D_{Axis} metric rather than on one neighborhood size.

Tokenizer identity. SHA-256 hashes are identical across all runs for the tokenizer vocabulary, configuration, and special-token map. A condition-specific tokenizer therefore cannot explain the result.

Channel-slice diagnostic. We divide the hidden dimension into contiguous head-width channel slices and compute the metric within each slice. Figure 15 shows a distributed middle-layer pattern rather than one isolated channel group. Attention-head output interventions offer a complementary direction for future work.

A.4.5 Differences Across Categories and Languages

Category and word results. The following analyses show which probes contribute most to the contextual-state difference, with signed examples and breakdowns by category and word.

Additional languages. Figure 19 completes the per-language view for the six languages outside the earlier Chinese and French analysis. Panel (a) compares raw target-language differences in contextual states and token embeddings, then shows how separating English documents changes each metric. Panel (b) places the six languages against coarse script and family indicators; the plotted points show no clear monotonic relationship.

Question	Best evidence	Pattern to look for
Does the difference exceed ordinary training variation?	Figure 9 and Table 11	Contextual-state points usually lie farther above the English-only seed reference than token-embedding points.
Could English exposure, training-step count, or repeated documents explain it?	Table 1, Figures 3 and 4	The contextual difference remains under matched exposure, matched total training, and separate English documents.
Is the result a consequence of the fitted coordinate map?	Table 2, Figure 14, and Table 13	Contextual-state differences remain after contextual-anchor and affine alignment and after changing the anchor subset.
When does the difference appear?	Figures 6 and 13	It is visible at the earliest saved checkpoint, becomes largest in middle layers, and survives vector-length controls.
Is it confined to the semantic probe categories?	Table 10 and Figure 16	Physical-vocabulary controls also differ, while category magnitudes vary. The result is broader than a uniquely cultural effect.

Table 9: **A reader's guide to the robustness evidence.** Each row links a possible alternative explanation to the display that tests it and states the visual pattern needed to evaluate the claim.

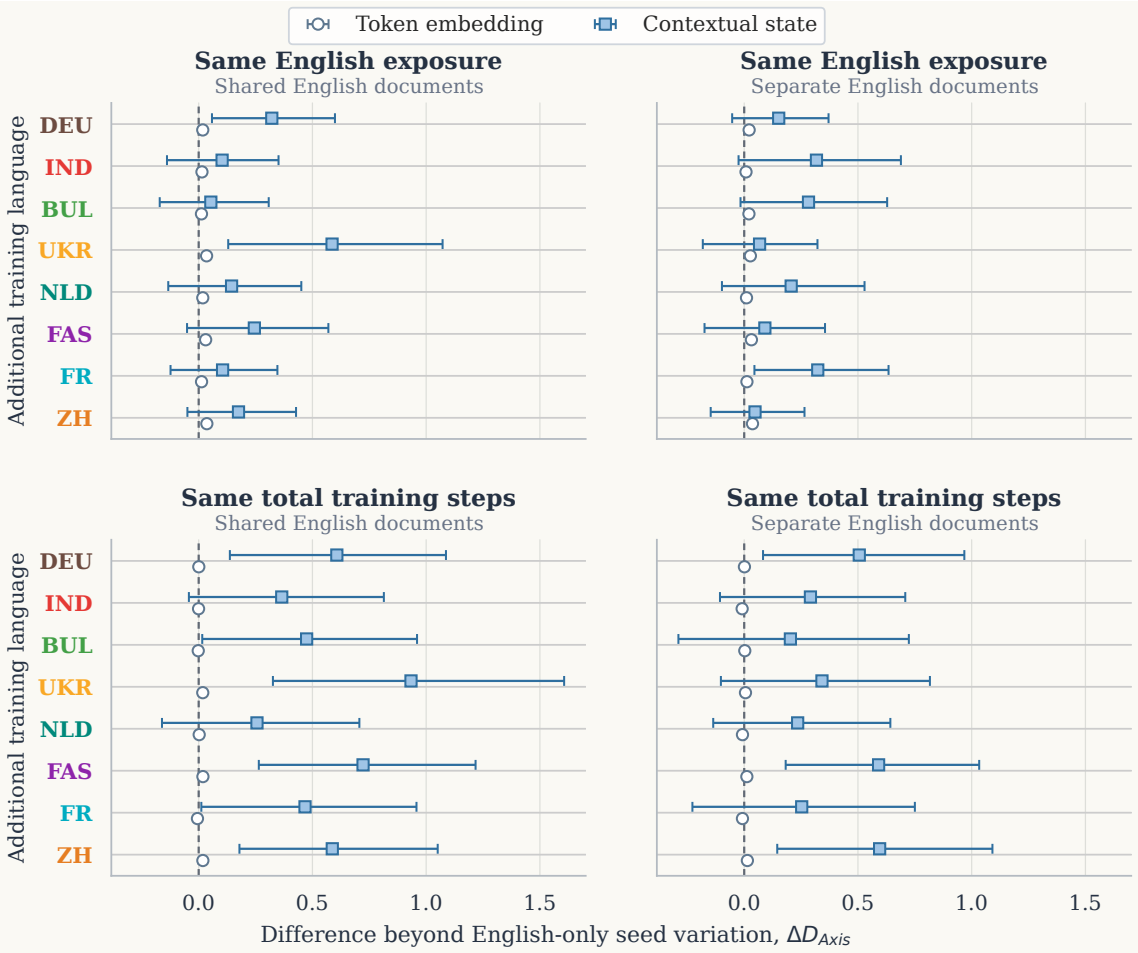


Figure 9: **Language-level uncertainty under all four training controls.** Circles are token embeddings and squares are contextual states; horizontal bars are 95% bootstrap intervals. The top row holds English exposure fixed, so both models receive 1,500 English training steps. The bottom row holds total training fixed at 3,000 steps. The right column replaces shared English documents with non-overlapping sets. Values are differences beyond the English-only seed mean.

Representation	Group	ΔD_{NN}	ΔD_{Struct}	ΔD_{Axis}
Token embedding	Cultural probes	0.029	0.014	0.014
	Negative controls	0.328	0.017	0.075
Contextual state	Cultural probes	0.015	0.042	0.386
	Negative controls	0.165	0.034	0.483

Table 10: **Physical-vocabulary controls show that the difference extends beyond the semantic probe categories.** The 100 controls are common physical-vocabulary terms. Their seed-centered disagreement is comparable to or larger than the primary probe summary, which places the result at the level of general representation change rather than a uniquely cultural effect.

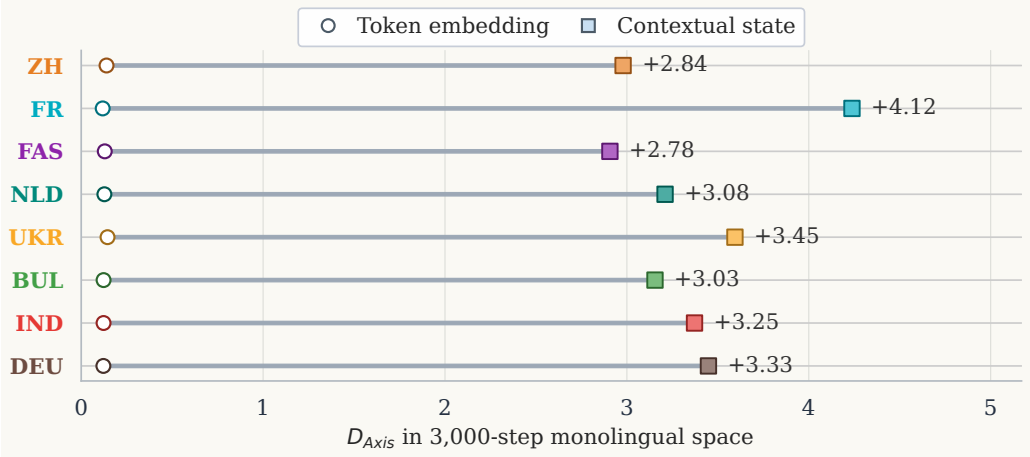


Figure 10: **The representation-level ordering also holds in the additional-language spaces under matched total training.** Each monolingual reference receives 3,000 steps in the additional language, while its paired bilingual model receives 1,500 steps in English and 1,500 in that language. Contextual states differ more than token embeddings for all eight languages.

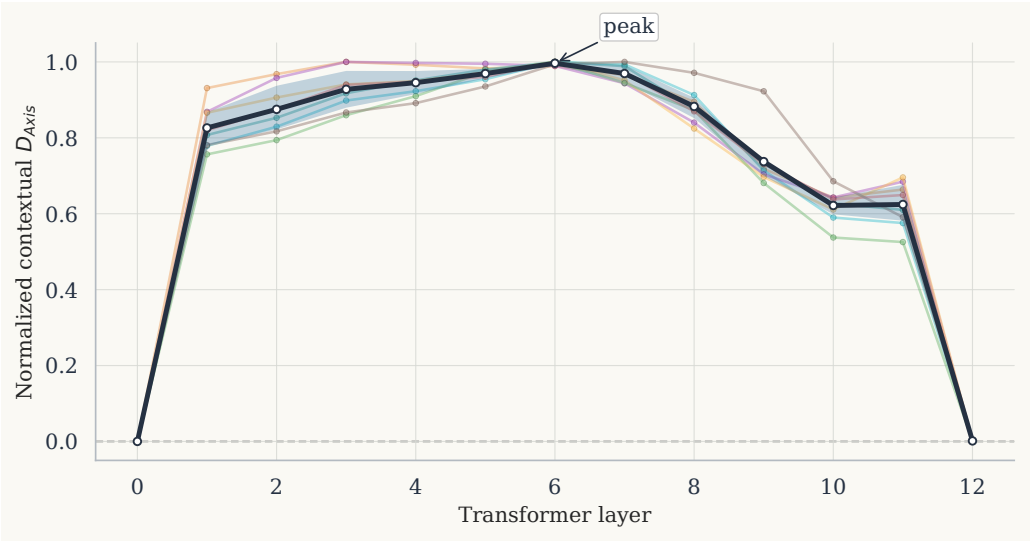


Figure 11: **The middle-layer peak remains when total training is matched.** The English-only and bilingual models each receive 3,000 training steps. Thin lines show the eight additional-language conditions, and the dark line is their mean.

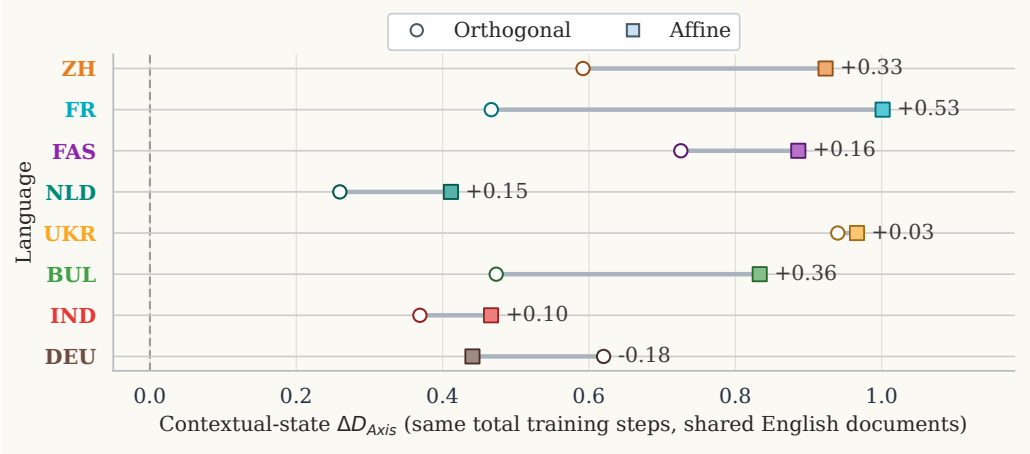


Figure 12: **Alternative maps preserve the matched-total-training result.** Lines connect the orthogonal and affine contextual-state values for each additional training language when both models receive 3,000 training steps. The map changes the numerical scale but does not reduce the result to the English-only seed reference.

Setting	Representation	Mean raw D_{Axis}	Std. dev.	English seed pairs
Same English exposure	Contextual state	2.258	0.182	6
Same English exposure	Token embedding	0.113	0.002	6
Same total training steps	Contextual state	3.204	0.295	6
Same total training steps	Token embedding	0.150	0.002	6

Table 11: **English-only runs define the reference for ordinary training variation.** Raw D_{Axis} is summarized across six ordered comparisons, corresponding to three unique seed pairs, for each training length and representation. The matching mean is subtracted from each bilingual comparison reported as ΔD_{Axis}.

Representation	Matched-axis D_{Axis}	Held-out-axis D_{Axis}	Held-out minus matched
Contextual state	0.261	0.390	0.129
Token embedding	0.018	0.014	-0.004

Table 12: **The contextual result transfers to held-out axis groups.** The axes used for the matched column and those reserved for the held-out column are disjoint. Contextual-state differences remain elevated for both groups, so one handcrafted theme does not drive the aggregate pattern.

	Held-out mean D_{Axis}		Alignment residual per anchor	
Setting	Token embedding	Contextual state	Token embedding	Contextual state
Same English exposure	0.135	2.439	0.069 → 0.038	0.378 → 0.170
Same total training steps	0.155	3.552	0.081 → 0.044	0.495 → 0.217

Table 13: **The representation-level pattern is stable as the alignment set grows from 500 to 3,000 words.** Held-out D_{Axis} is averaged over eight additional languages and five random anchor draws. A single value means the rounded result is unchanged across subset sizes. The arrows show that alignment fit improves with more anchors while contextual-state differences remain much larger than token-embedding differences.

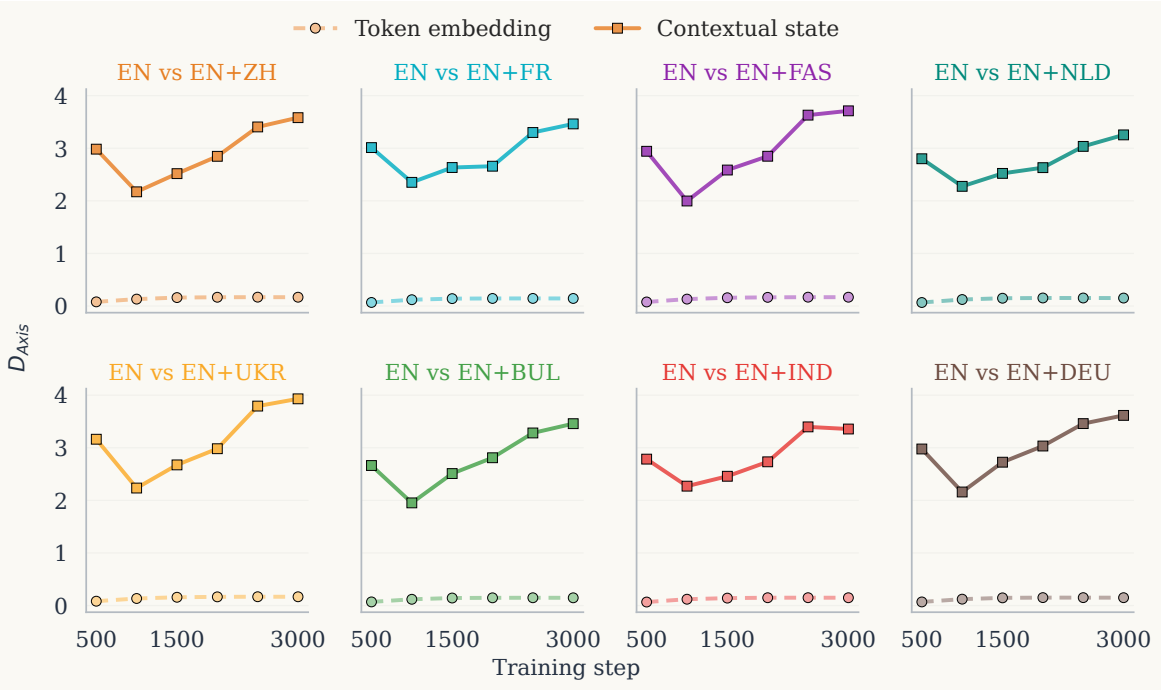


Figure 13: **Contextual states differ more than token embeddings throughout training.** Each panel traces one additional-language condition at 500-step intervals when both models see 1,500 English training steps and reuse the same English documents. The difference is present at the earliest saved checkpoint and remains through 3,000 steps.

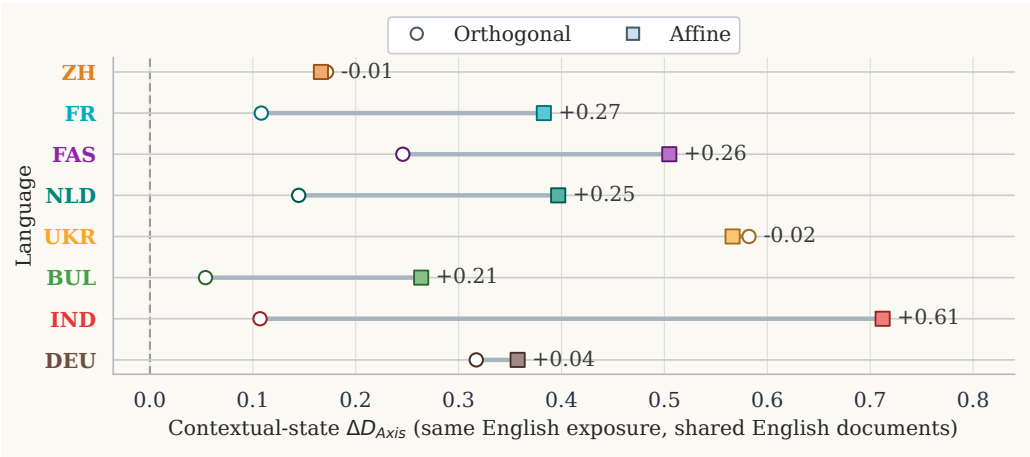


Figure 14: **The alignment method changes values, not the main result.** Lines connect orthogonal and affine contextual-state values for each additional training language when both models see 1,500 English training steps and share the English documents.

	Token embedding			Contextual state		
k	English-only	Bilingual	Difference	English-only	Bilingual	Difference
5	0.568	0.590	0.023	0.807	0.827	0.019
10	0.646	0.668	0.022	0.796	0.809	0.013
25	0.613	0.642	0.029	0.772	0.787	0.015
50	0.610	0.653	0.043	0.781	0.804	0.023
100	0.779	0.803	0.024	0.778	0.803	0.025

Table 14: **The complementary neighbor metric remains above seed variation across neighborhood sizes.** English-only values summarize the six ordered same-language comparisons; bilingual values summarize the sixteen shared-document comparisons across the matched-exposure and matched-total-training settings. The last column in each representation block is their difference.

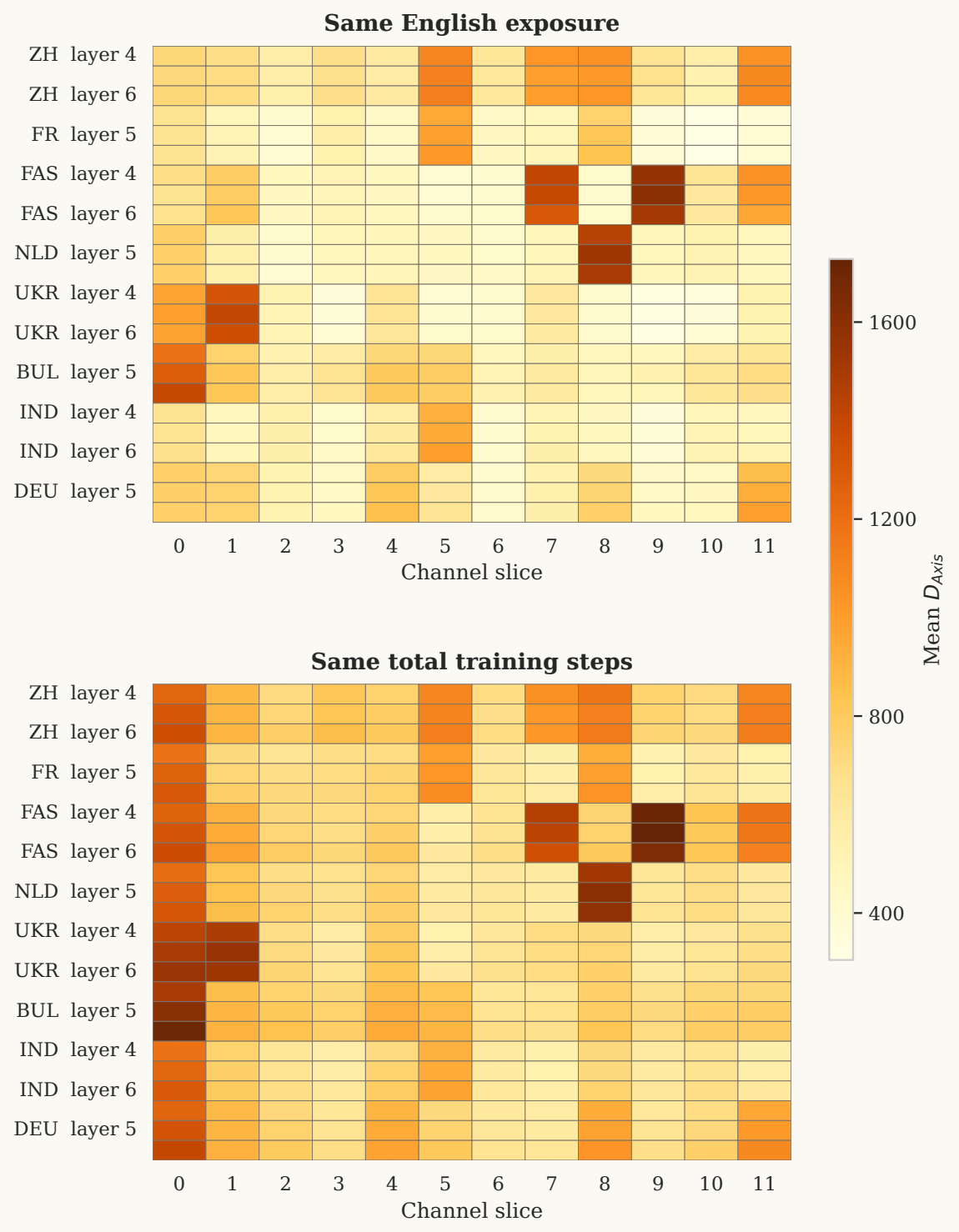


Figure 15: **The middle-layer difference is distributed across channel slices.** Each cell reports mean D_{Axis} for one contiguous head-width slice of the hidden dimension.

Test block	Slice	n	Mean diff.	One-sided p
Main EN-centered gap	Same English exposure	16	0.181	1.53e-05
Main EN-centered gap	Same total training steps	16	0.464	1.53e-05
Main EN-centered gap	All	32	0.322	$< 10^{-6}$
Anchor-subset reruns	Same English exposure	160	2.304	$< 10^{-6}$
Anchor-subset reruns	Same total training steps	160	3.397	$< 10^{-6}$
Anchor-subset reruns	All	320	2.850	$< 10^{-6}$

Table 15: **Aggregate tests confirm the representation-level gap.** The two training settings state which quantity is held fixed.

Highest contextual divergence			Lowest contextual divergence		
Rank	Word	Mean D_{NN}	Rank	Word	Mean D_{NN}
1	vegan	0.999	1	extended mother	0.323
2	stereotype	0.997	2	public democracy	0.333
3	bamboo	0.997	3	extended father	0.338
4	integrity	0.996	4	extended son	0.341
5	carnival	0.996	5	extended daughter	0.361
6	jewelry	0.996	6	sacred prayer	0.362
7	espresso	0.996	7	public ministry	0.373
8	lineage	0.996	8	sacred mosque	0.373
9	homeland	0.996	9	local majority	0.381
10	governance	0.995	10	sacred curse	0.384

Table 16: **Contextual neighbor disagreement varies substantially across probes.**

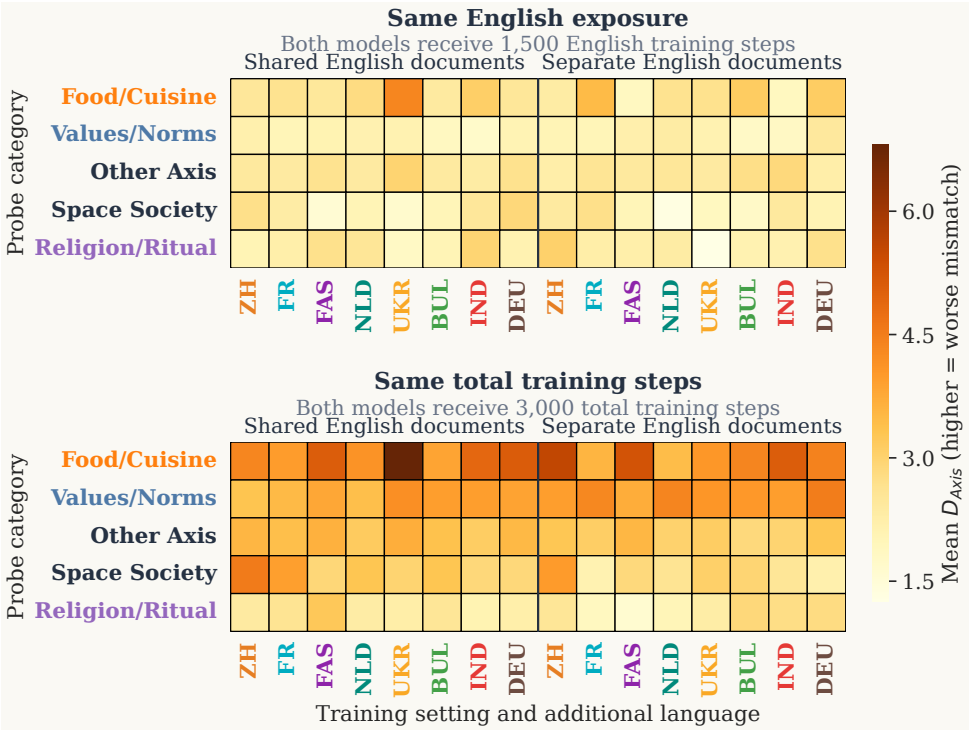


Figure 16: **Contextual differences vary across probe categories.** `Food/Cuisine`, `Symbols/Colors`, and `Social Identity` contribute disproportionately, while the physical-vocabulary controls in Table 10 place the effect at the level of general representation change.

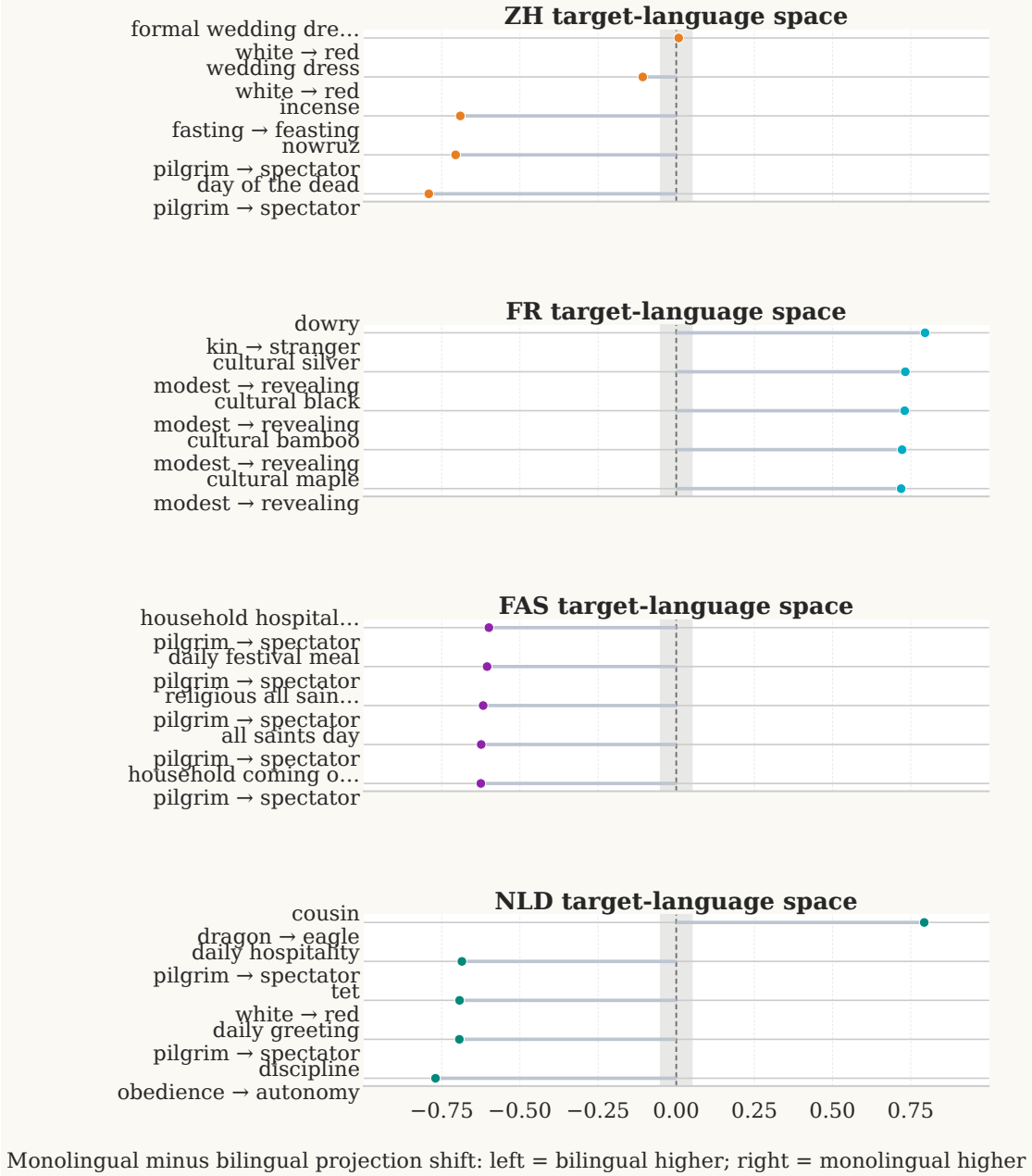


Figure 17: **Largest signed shifts in the Chinese, French, Persian, and Dutch target-language spaces.** Each panel names the probe and semantic contrast and reports the monolingual-minus-bilingual projection difference.

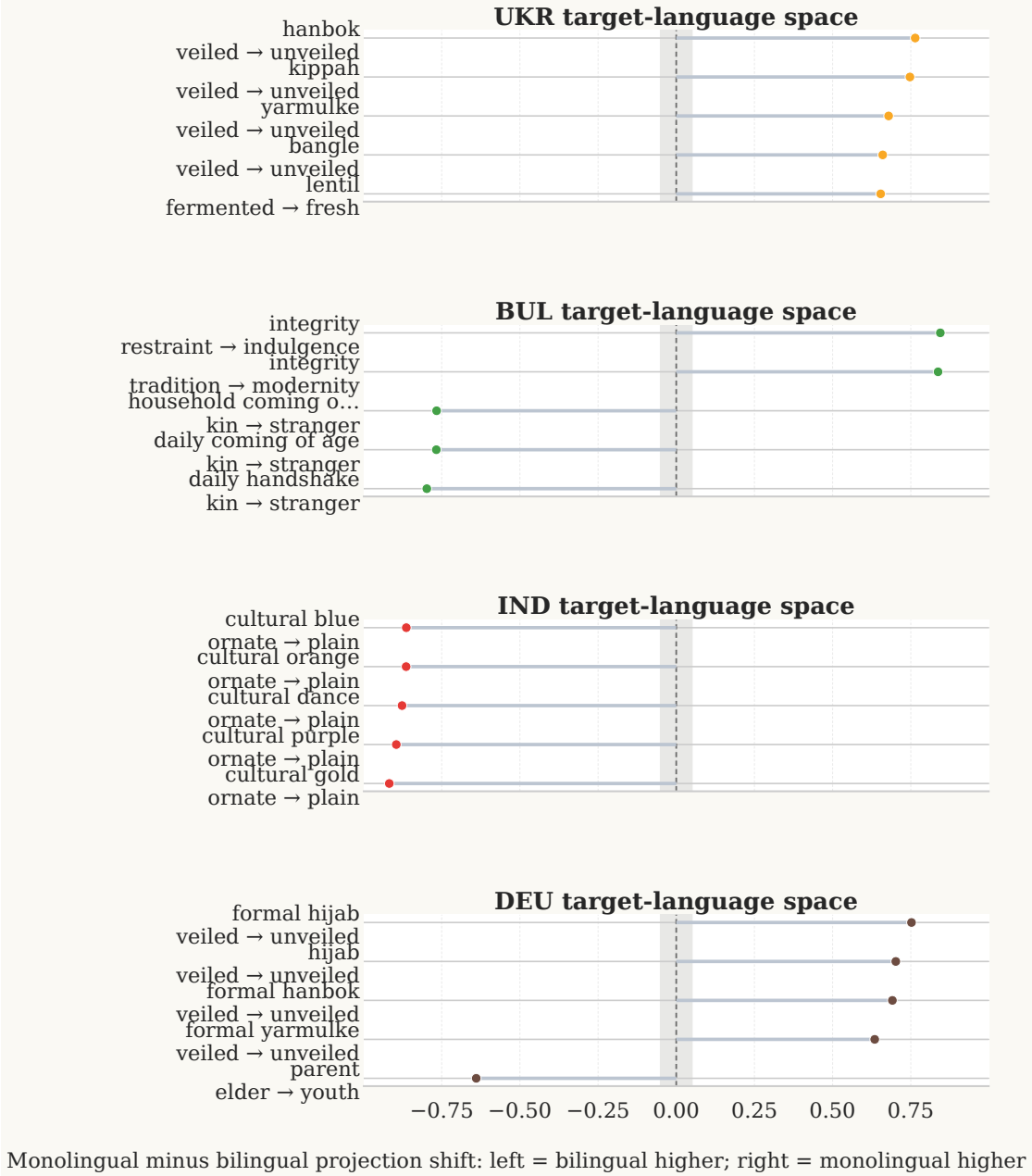
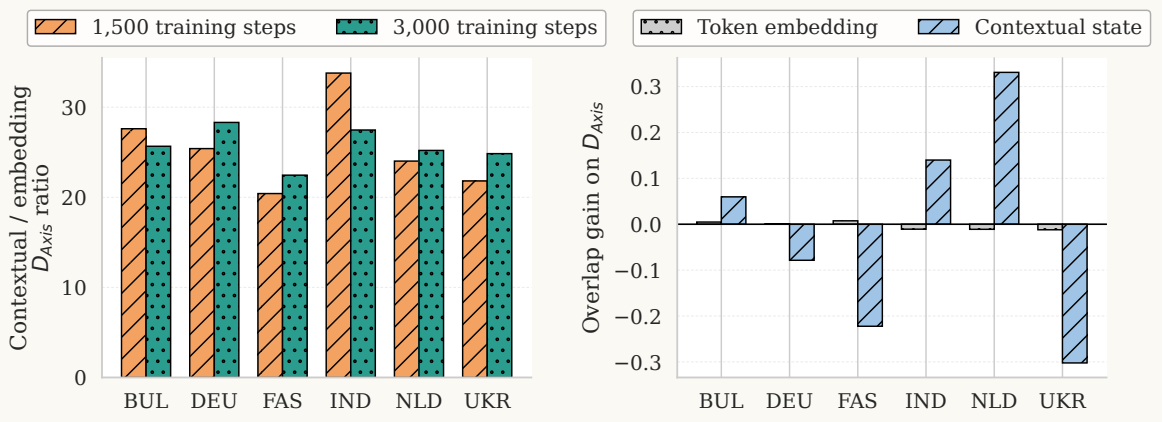
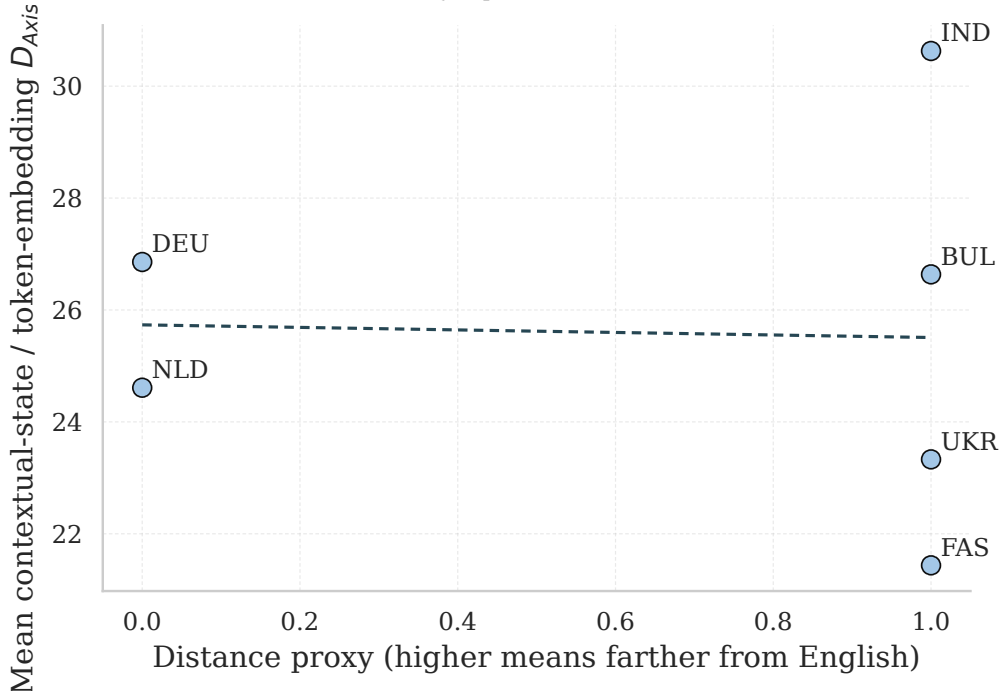


Figure 18: **Largest signed shifts in the Ukrainian, Bulgarian, Indonesian, and German target-language spaces.** Each panel names the probe and semantic contrast and reports the monolingual-minus-bilingual projection difference.



(a) Target-space representation ratios and the change after replacing shared English documents with separate sets. The two monolingual references receive 1,500 and 3,000 training steps.



(b) Exploratory comparison of six languages using coarse script and family indicators; the points show no monotonic relation.

Figure 19: **Per-language patterns for six additional conditions preserve the representation-level ordering.** Contextual-state differences remain larger than token-embedding differences, while the exploratory typology comparison shows no clear monotonic relationship.